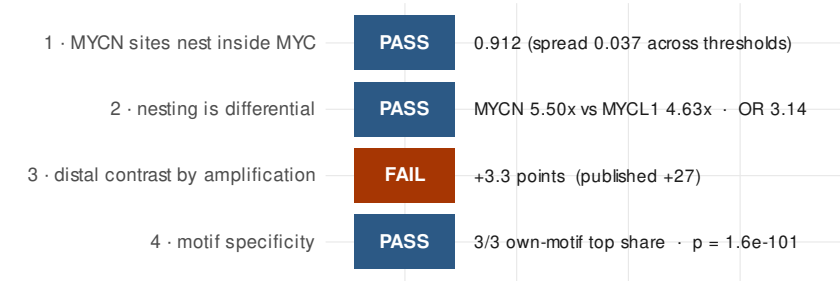


The **paralog-resolved regulatory layer is internally sound, with one criterion failed**

Cell-line chromatin, 10 SCLC lines. This is the evidence that the programmes tested in Figure 1 were real before they were tested in tumours; the criterion that failed is shown rather than omitted.

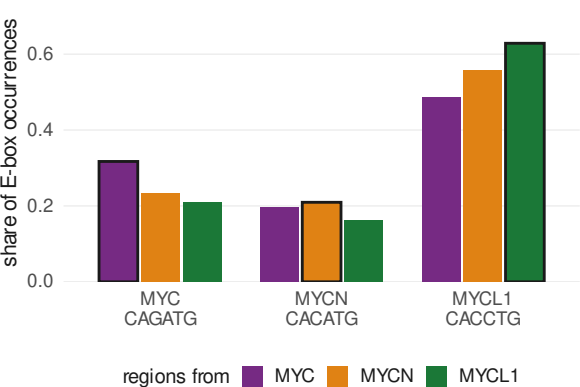
A · The M5 gate, including where it failed

cell-line chromatin, 3 of 4 criteria pass



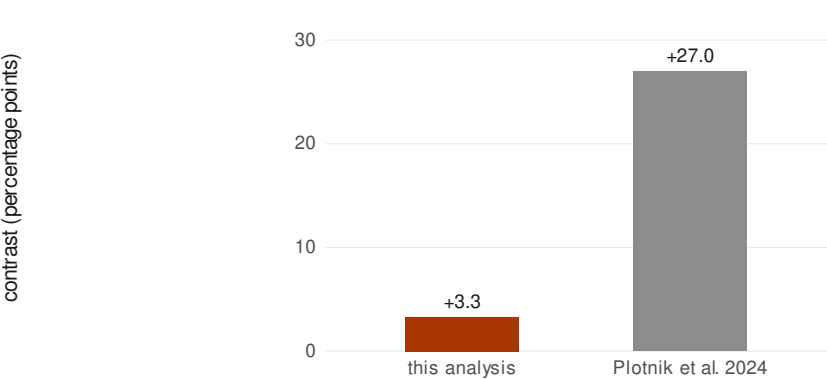
B · Each motif peaks in its own regions

compositional share within each region set;
outlined bar is the own-set bar — highest in 3/3



C · Criterion 3 fails on magnitude, not direction

distal-fraction contrast, MYC-amplified minus MYC-expressing
amplified: H524 53.7%, H211 41.2% · expressing: H1048 41.2%, SHP77 47.2%



D · Regulons survive holding out a line

leave-one-line-out AUC, 8 of 9 at or above the dashed 0.70 threshold

